

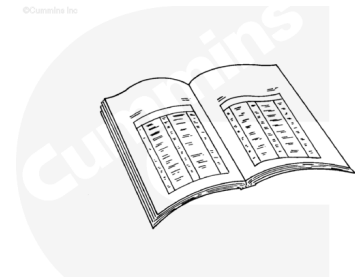
Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

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ck800wa

⚠ WARNING ⚠

Coolant is toxic. Keep away from small children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

- Attach tags to all hoses, lines, linkages, and electrical connections, as they are removed, to identify location and aid the installation process.
- Disconnect the battery cables. Refer to Procedure 013-009 (Battery Cables and Connections) in Section 13. (</qs3/pubsys2/xml/en/procedures/99/99-013-009.html>)
- Drain the engine coolant. Refer to Procedure 008-018 (Cooling System) in Section 8. (</qs3/pubsys2/xml/en/procedures/28/28-008-018-tr.html>)
- Drain the lubricating oil. Refer to Procedure 007-037 (Lubricating Oil System) in Section 7. (</qs3/pubsys2/xml/en/procedures/28/28-007-037-tr.html>)
- Attach tags to all hoses, lines, linkages, and electrical connections, as they are removed, to identify location and aid the installation process.
- Disconnect the batteries. Refer to Procedure 013-009 (Battery Cables and Connections) in Section 13. (</qs3/pubsys2/xml/en/procedures/99/99-013-009.html>)
- Disconnect the air/fuel supply to the air starter if equipped. Refer to Procedure 012-022 (Air Starting Motor) in Section 12. (</qs3/pubsys2/xml/en/procedures/28/28-012-022-tr-gc.html>)
- Drain the engine coolant. Refer to Procedure 008-018 (Cooling System) in Section 8. (</qs3/pubsys2/xml/en/procedures/28/28-008-018-tr-gc.html>)
- Drain the lubricating oil from the engine. Refer to Procedure 007-037 (Lubricating Oil System) in Section 7. (</qs3/pubsys2/xml/en/procedures/28/28-007-037-tr-gc.html>)
- Disconnect the starter cable, engine ground straps, cab or chassis to engine hoses, tubing, electrical wires, and hydraulic lines.
- Disconnect all hoses, tubing, electrical wires, and hydraulic lines, if equipped.
- Disconnect the intake and exhaust system pipes. Refer to the OEM installation instructions.

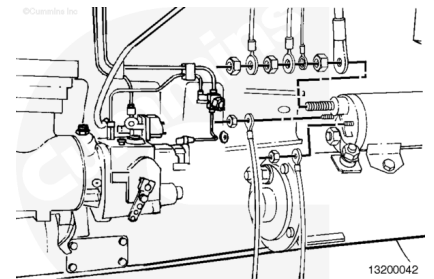
- Disconnect the drive units from the flywheel housing and flywheel. Refer to the OEM service manual.
- Remove air conditioning lines, if needed. Refer to the OEM service manual.
- Remove all chassis components necessary to remove the engine from the equipment. Refer to the OEM service manual.

Remove

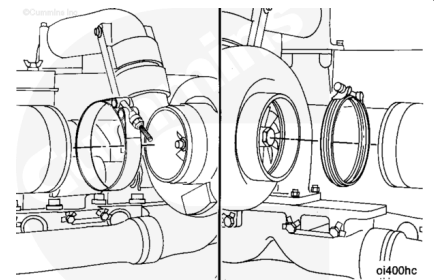
Note : The engine illustrations in this procedure could be different for a specific application, but the removal steps are the same regardless of engine type.

Disconnect the starter cable, engine ground straps, cab or chassis to engine hoses, tubing, electrical wires, and hydraulic lines.

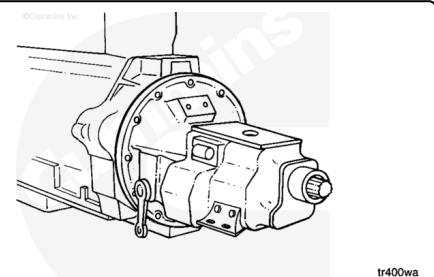
Disconnect all hoses, tubing, electrical wires, and hydraulic lines, if equipped.



Disconnect the intake and exhaust system pipes.



Disconnect the drive units from the flywheel housing and flywheel.



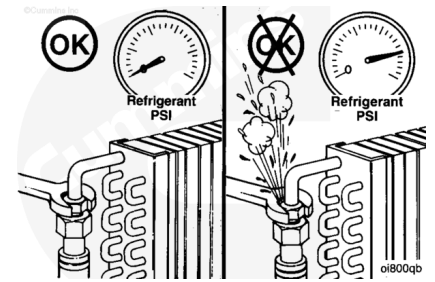
⚠ WARNING ⚠

If a liquid refrigerant system (air conditioning) is used, wear eye and face protection, and wrap a cloth around the fittings before removing. Liquid refrigerant can cause serious eye

and skin injuries.

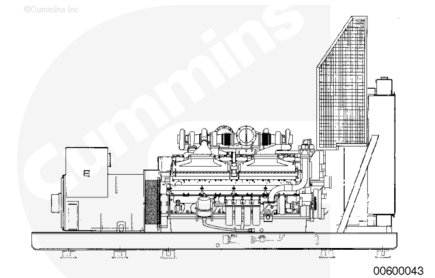
⚠ WARNING ⚠

To protect the environment, liquid refrigerant systems must be properly emptied and filled using equipment that prevents the release of refrigerant gas into the atmosphere. Federal law requires capturing and recycling the refrigerant.



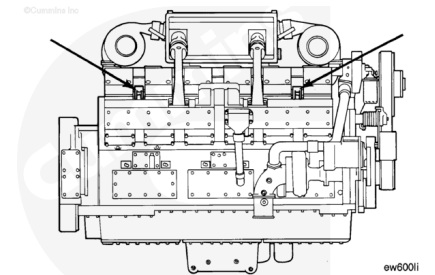
Remove air conditioning lines if needed.

Remove all chassis components necessary to remove the engine from the equipment.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Note : Before lifting the engine, check the four engine lifting brackets. Be sure each bracket contains a spot-face under the hardened flat washer. If the bracket is not spot-faced, remove the capscrew. Check for rust and corrosion. If the capscrew has rust or corrosion, replace the capscrew before lifting the engine. Refer to Procedure 016-001 (Engine Lifting Brackets) in Section 16. (</qs3/pubsys2/xml/en/procedures/28/28-016-001-tr.html>) Refer to Procedure 016-001 (Engine Lifting Brackets) in Section 16. (</qs3/pubsys2/xml/en/procedures/28/28-016-001-tr-gc.html>)

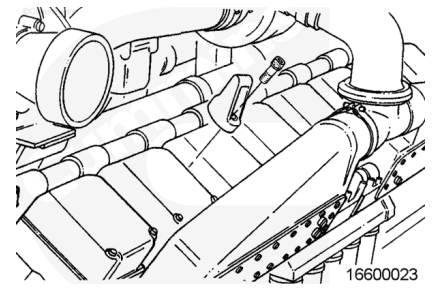
KTA38GC, K38 and QSK38 engines have four lifting brackets, two on each bank. The lifting brackets for the K38 are installed between cylinder numbers 1 and 2, and 5 and 6.

The lifting brackets for the K50 and QSK50 are installed between cylinder numbers 2 and 3, and 6 and 7.

Remove the lifting brackets. Check for cracks, rust and corrosion. Install lifting brackets. Refer to Procedure 016-001 (Engine Lifting Brackets) in Section 16.

(/qs3/pubsys2/xml/en/procedures/28/28-016-001-tr.html)

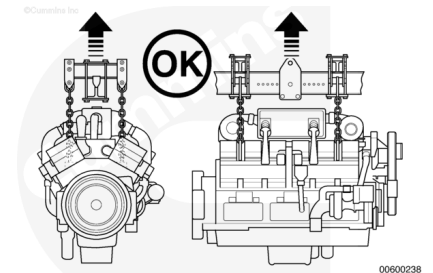
Refer to Procedure 016-001 (Engine Lifting Brackets) in Section 16. (/qs3/pubsys2/xml/en/procedures/28/28-016-001-tr-gc.html)



The engine lifting fixtures, Part Number 3163264, are designed to lift the engine **only**. Any additional components attached (including to the drive end, non-drive end and/or chassis) **must** be removed.

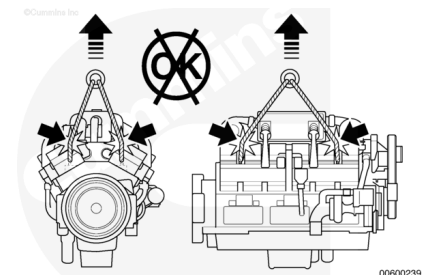
The lifting force **must** be in a straight vertical line with the lifting brackets in both directions.

Engine lifting fixture, Part Number 3163264, is available from Cummins Inc. This fixture is available to lift the KTA38GC, K38, K50, QSK38, QSK50 assembled engines.

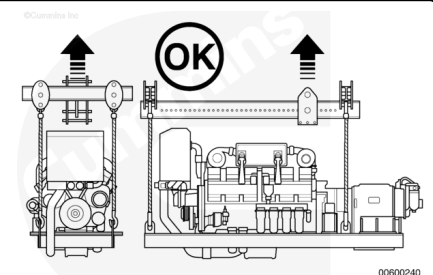


⚠ CAUTION ⚠

Do not attempt to lift the engine with an apparatus that allows side-loading of the lifting brackets, or allows the chain or cable to apply force to other parts of the engine. The lifting brackets or parts can break, causing the engine to fall and be damaged.

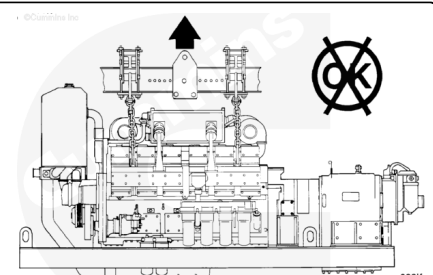


Always use the lifting brackets that are connected to the base rails when drive units or a radiator is mounted on the base rails with the engine.



⚠ CAUTION ⚠

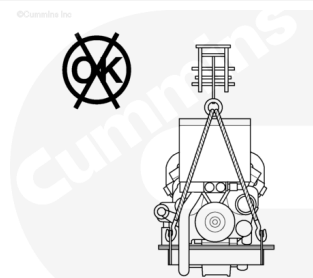
Never use the engine lifting brackets or the engine lifting fixture, Part Number 3163264, to lift an engine when drive units or a radiator is mounted on common base rails. Never use the lifting bracket on the block to lift an engine that is



mounted on base rails when there is a driven or radiator also mounted on the base rails. The lifting brackets or parts can break, causing the engine to fall. Engine damage can result.

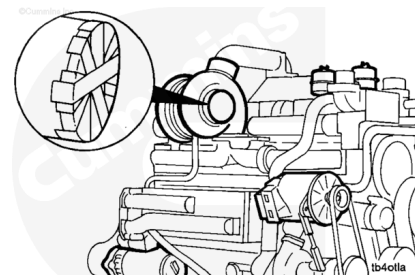
⚠ CAUTION ⚠

Do not attempt to lift the engine with an apparatus that allows side-loading of the lifting brackets, or allows the chain or cable to apply force to other parts of the engine. The lifting brackets or parts can break, causing the engine to fall and be damaged.



Finishing Steps

Cover all engine openings to prevent dirt and debris from entering the engine.



Remove all remaining accessories and brackets that will be used on the replacement engine.

